IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies:

Data was collected via SpaceX REST API and web scraping. Data was wrangled and cleaned to create binary landing outcome labels. Exploratory data analysis was performed using visualization and SQL. Interactive maps and dashboards were built using Folium and Plotly Dash. Four classification models were trained and evaluated using GridSearchCV.

- Summary of all results

KSC LC-39A has the highest success rate (76.9%). Launch success improved significantly over time. Lower payloads correlate with higher success. Decision Tree classifier achieved the best test accuracy of 88.9%.

Introduction

- Project background and context

SpaceX Falcon 9 launches cost $62M vs competitors' $165M+ largely due to first stage reusability. Predicting whether the first stage will successfully land allows estimation of launch cost — valuable for companies competing with SpaceX.

- Problems you want to find answers

- Can we predict Falcon 9 first stage landing success?
- Which launch sites have the highest success rates?
- What payload ranges and orbits correlate with successful landings?
- Which machine learning model best predicts landing outcome?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Describe how data was collected
- Perform data wrangling
  - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - How to build, tune, evaluate classification models

Data Collection

- Data was collected using two methods: SpaceX REST API for structured launch records, and web scraping Wikipedia for historical Falcon 9 launch tables.

- Both datasets were combined to support the full analysis pipeline.

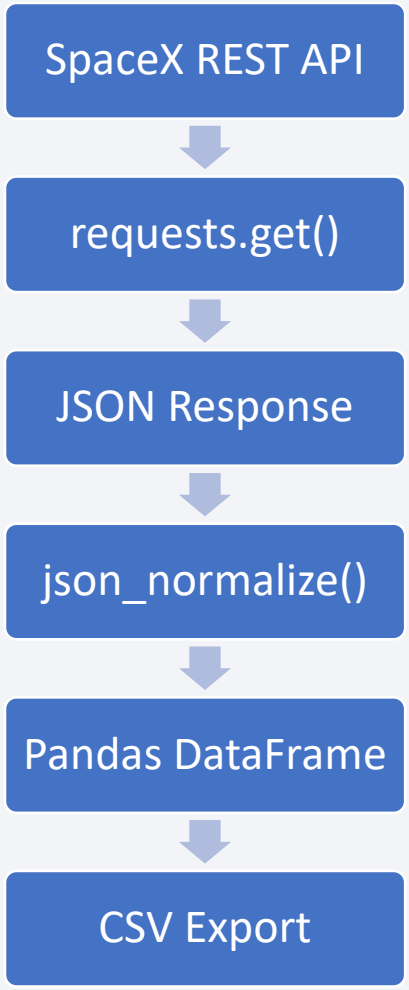
  SpaceX REST API → DataFrame

  Wikipedia Scraping → DataFrame

Data Collection – SpaceX API

- SpaceX REST API was called using Python requests library. JSON responses were parsed and normalized into a Pandas dataframe containing flight number, date, booster version, payload mass, orbit, launch site, and landing outcome.

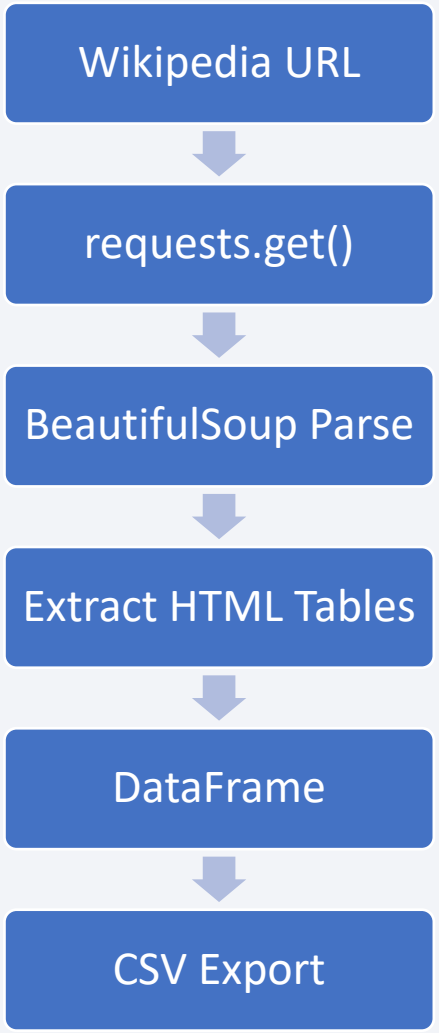
- https://github.com/KatyaMak/IBM_Capstone/blob/main/10_Lab01_Data_Collection.ipynb



Data Collection - Scraping

- Historical Falcon 9 launch data was scraped from Wikipedia using Python BeautifulSoup library. HTML tables were parsed, missing values handled defensively, and data structured into a Pandas dataframe.

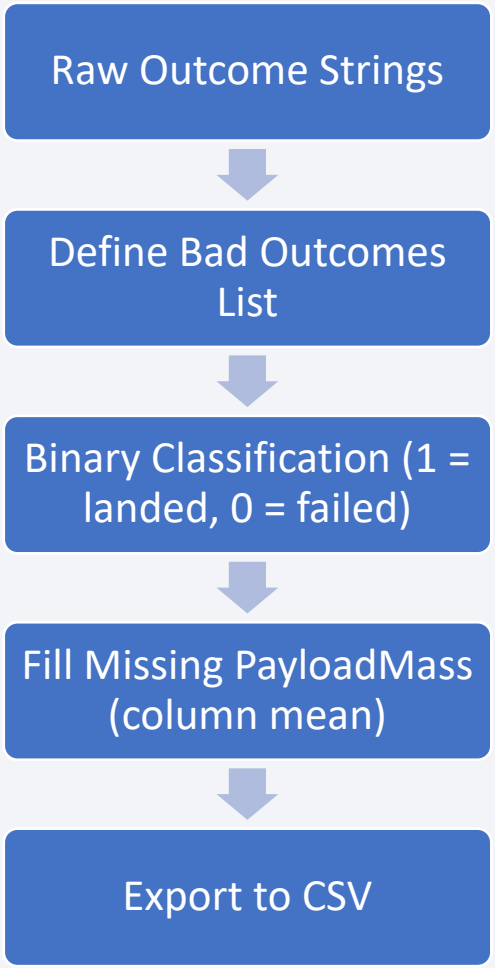
- https://github.com/KatyaMak/IBM_Capstone/blob/main/10_Lab02_WebScraping.ipynb



Data Wrangling

- Raw landing outcome strings (True RTLS, False Ocean, True ASDS, etc.) were converted into binary training labels: 1 = successful landing, 0 = unsuccessful. Missing PayloadMass values were filled with column mean.
- Overall success rate: 66.7%.

- **GitHub URL:** https://github.com/KatyaMak/IBM_Capstone/blob/main/10_Lab03_Data_Wrangling.ipynb



EDA with Data Visualization

Six scatter and line charts were plotted using Seaborn and Matplotlib to explore relationships between launch variables and landing success:

- Flight Number vs Launch Site — to see which sites were used over time and their success patterns
- Payload Mass vs Launch Site — to identify payload capacity patterns per site
- Success Rate by Orbit Type — bar chart to compare landing success across orbit types
- Flight Number vs Orbit Type — to see if experience improved outcomes per orbit
- Payload Mass vs Orbit Type — to identify payload-outcome patterns per orbit
- Yearly Success Trend — line chart showing improvement in success rate over time (2013–2020)

- GitHub URL: https://github.com/KatyaMak/IBM_Capstone/blob/main/10_Lab05_EDA_Visualization.ipynb

EDA with SQL

SQL queries performed:

- Queried all unique launch sites from the dataset
- Retrieved launches with payload mass between 4000–6000 kg
- Identified total payload mass carried by NASA (CRS) missions
- Calculated average payload mass by booster version F9 v1.1
- Found date of first successful ground landing
- Listed boosters with successful drone ship landings with payload between 4000–6000 kg
- Counted total successful and failed mission outcomes
- Found all launch site names beginning with 'CCA'
- Retrieved 2015 drone ship failure records by month and booster
- Ranked landing outcomes between 2010-06-04 and 2017-03-20 in descending order

https://github.com/KatyaMak/IBM_Capstone/blob/main/10_Lab04_EDA_SQL.ipynb

Build an Interactive Map with Folium

- Circles + text markers — added for each of the 4 launch sites to show their geographic location on a global map. Labels display site names using DivIcon.
- MarkerCluster — added all individual launches as clustered markers, color-coded green (success) and red (failure) to visualize landing outcomes per site.
- MousePosition plugin — enabled real-time coordinate display while hovering over the map, used to identify proximity coordinates.
- Distance markers + PolyLines — added labeled markers and connecting lines to show distance from CCAFS launch site to nearest coastline (under 1 km), railway (1.23 km), highway (9.29 km), and city — Cocoa Beach (24+ km).

These objects reveal that all launch sites share the same geographic pattern: coastal location for safety, rail access for logistics, and distance from populated areas.

- GitHub URL:

https://github.com/KatyaMak/IBM_Capstone/blob/main/10_Lab06_Interactive_Visualization.ipynb

Build a Dashboard with Plotly Dash

Plots/graphs and interactions added to a dashboard:

- Dropdown menu — allows selection of individual launch site or all sites, enabling focused analysis per location
- Pie chart — displays either total success count across all sites, or success vs failure ratio for selected site. Added to quickly compare site performance
- Range slider — filters payload mass from 0–10,000 kg, allowing exploration of how payload weight correlates with outcome
- Scatter plot — shows Payload Mass vs Launch Outcome, color-coded by Booster Version Category. Added to reveal which booster versions and payload ranges perform best

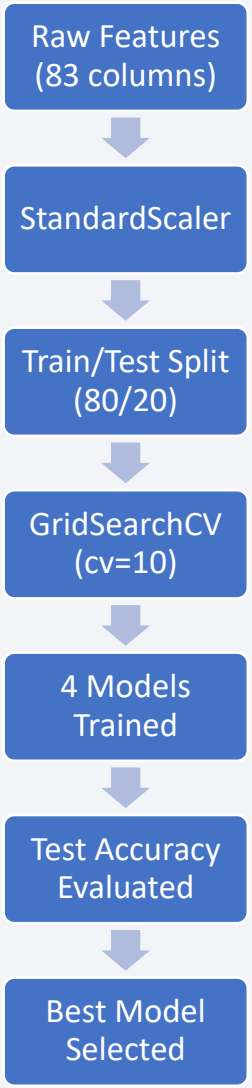
GitHub URL: https://github.com/KatyaMak/IBM_Capstone/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

Summary of process:

- Standardized features using StandardScaler
- Split data 80/20 into training and test sets
- Built four classification models: Logistic Regression, SVM, Decision Tree, KNN
- Used GridSearchCV with cv=10 to find best hyperparameters for each
- Evaluated each model on test data using accuracy score and confusion matrix

- GitHub URL: https://github.com/KatyaMak/IBM_Capstone/blob/main/10_Lab08_ML.ipynb



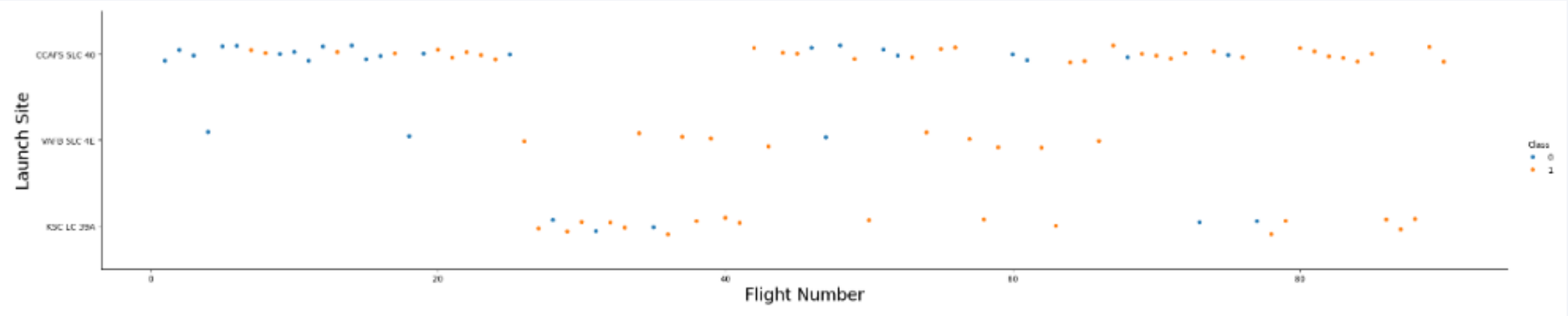
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

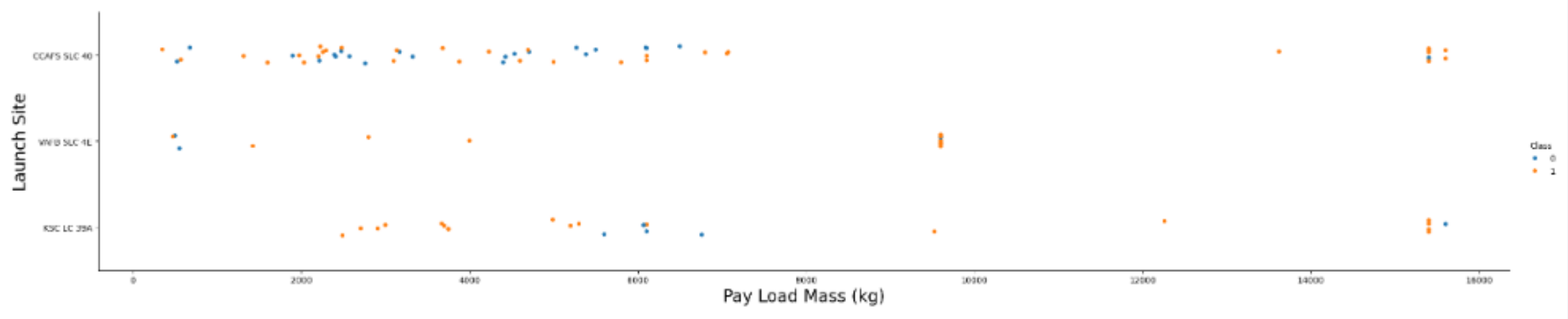
Insights drawn from EDA

Flight Number vs. Launch Site



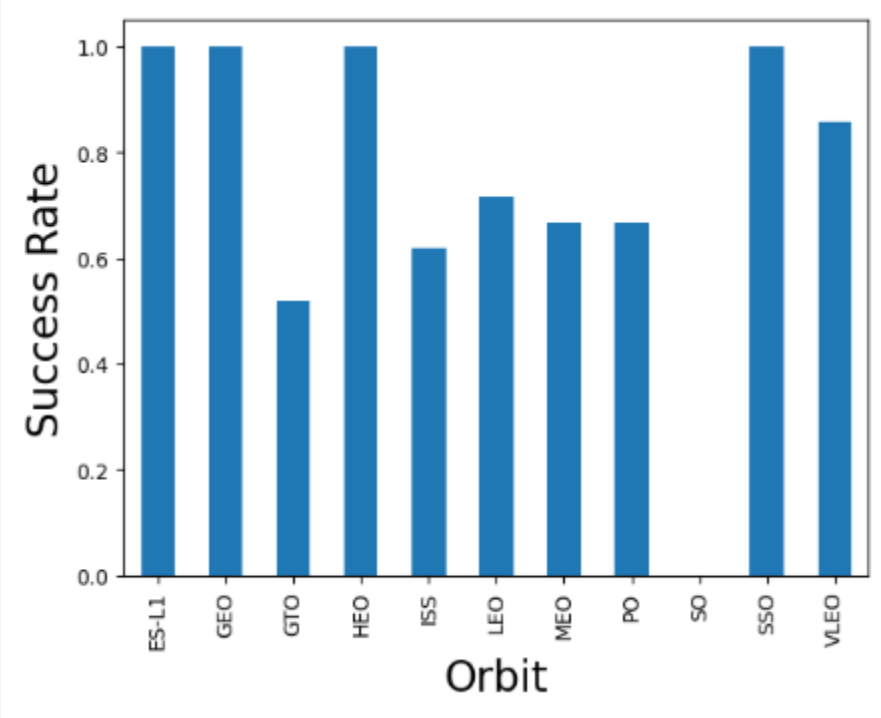
Later flight numbers show higher success rates across all sites. KSC LC-39A appears predominantly in later, more successful flights. CCAFS SLC 40 shows the full history including early failures.

Payload vs. Launch Site



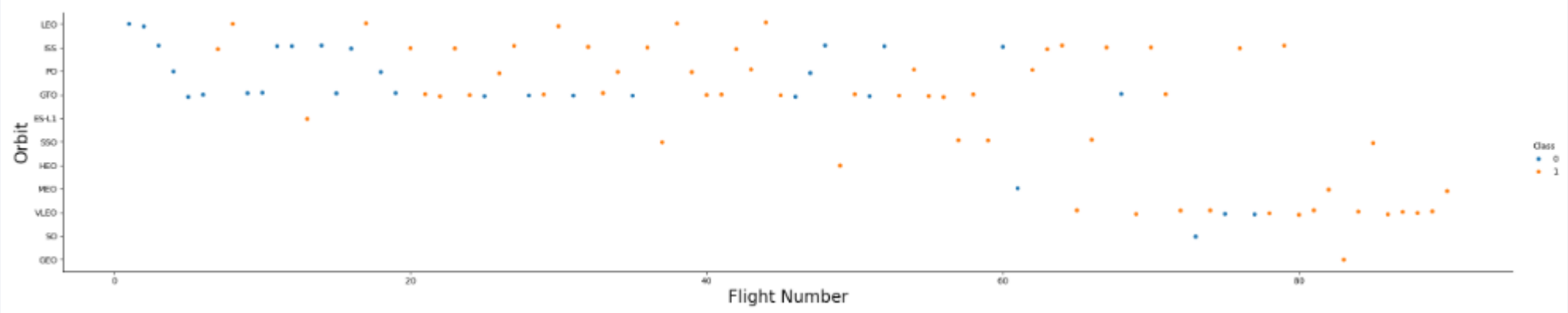
CCAFS SLC 40 handles the widest payload range. Heavier payloads show mixed outcomes across sites. VAFB SLC 4E tends toward lighter payloads.

Success Rate vs. Orbit Type



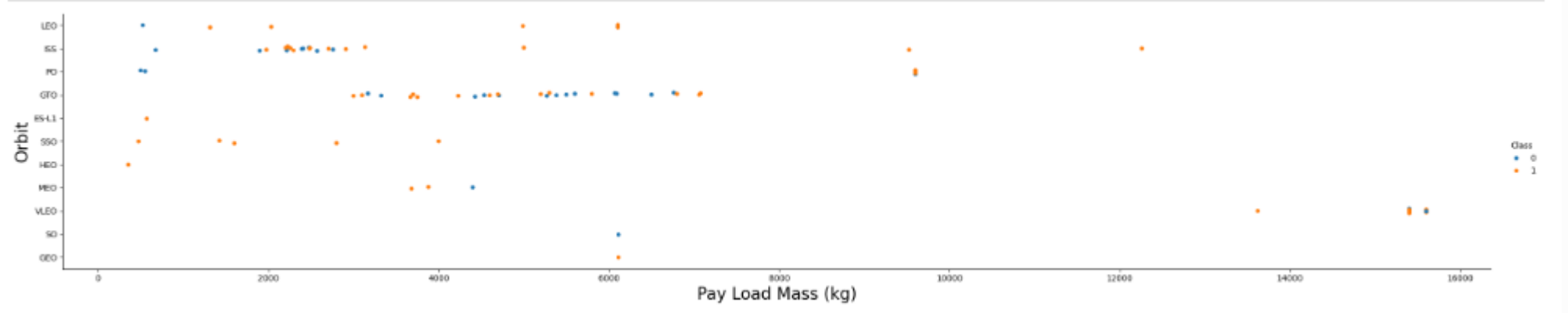
- ES-L1, GEO, HEO, and SSO orbits show 100% success rate. SO and GTO show lower rates. LEO and ISS show moderate success — these are the most frequently used orbits.

Flight Number vs. Orbit Type



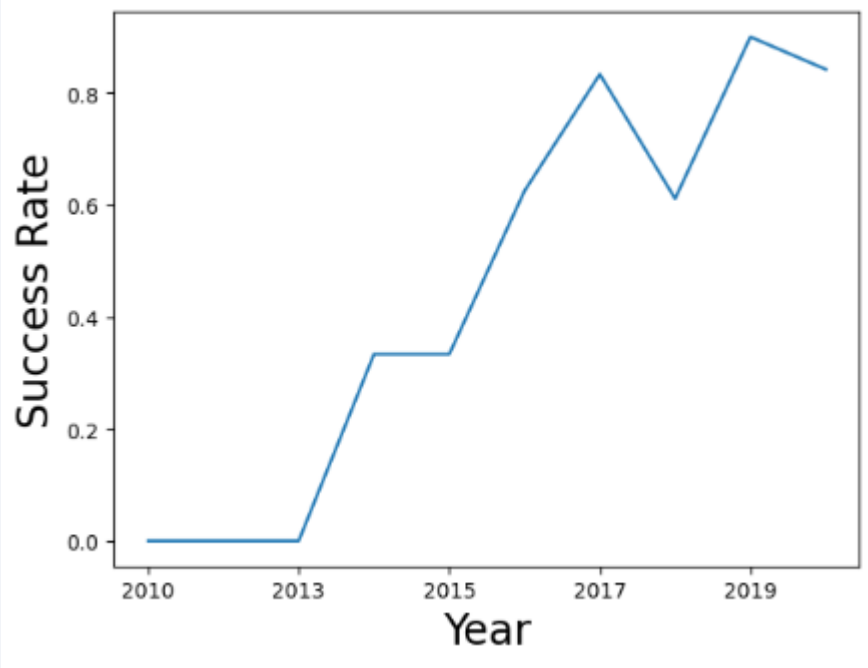
- LEO orbit shows clear improvement with higher flight numbers. GTO shows no clear relationship between experience and success. Polar orbits appear mostly in earlier flights.

Payload vs. Orbit Type



- GTO requires heavier payloads with mixed outcomes. LEO and ISS missions succeed across a wide payload range. Polar orbits tend toward lighter payloads.

Launch Success Yearly Trend



- Success rate increased steadily from 2013 to 2020. Notable jump after 2015 as SpaceX refined landing technology. By 2019-2020 success rate approached 100%.

All Launch Site Names

- The names of the unique launch sites:

- CCAFS LC-40,
- CCAFS SLC-40,
- KSC LC-39A,
- VAFB SLC-4E

- Four unique launch sites were used for Falcon 9 missions, all located in the United States.

Launch Site Names Begin with 'CCA'

- 5 records where launch sites begin with `CCA`:

- First 5 records from CCAFS LC-40 *CCAFS (Cape Canaveral Air Force Station)*

- CCAFS LC-40 *CCAFS (Cape Canaveral Air Force Station)* was the earliest and most frequently used launch site.

Total Payload Mass

- Calculate the total payload carried by boosters from NASA:

  45,596 kg

- NASA CRS missions carried the highest cumulative payload mass among all customers.

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

  2,928.4 kg

- Early booster version F9 v1.1 carried lighter payloads compared to later block versions.

First Successful Ground Landing Date

- The first successful landing outcome on ground pad:

  2015-12-22

- The first successful ground pad landing marked a historic milestone in reusable rocket technology.

Successful Drone Ship Landing with Payload between 4000 and 6000

- The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000:

  B1021, B1031, B1035, B1036

- These boosters successfully landed on drone ships while carrying heavier payloads.

Total Number of Successful and Failure Mission Outcomes

- The total number of successful and failure mission outcomes:

  Results: Success — 61, Failure — 29

- Overall mission success rate of approximately 67% across all recorded launches.

Boosters Carried Maximum Payload

- The names of the booster which have carried the maximum payload mass:

  B1048, B1049, B1051, B1056, B1060 (15,400 kg)

- These FT block 5 boosters carried the heaviest payloads in the dataset.

2015 Launch Records

- The failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015:

Results: Month 01 — F9 v1.1 B1012, CCAFS LC-40; Month 04 — F9 v1.1 B1015, CCAFS LC-40

- Both 2015 drone ship failures used early v1.1 boosters from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order:

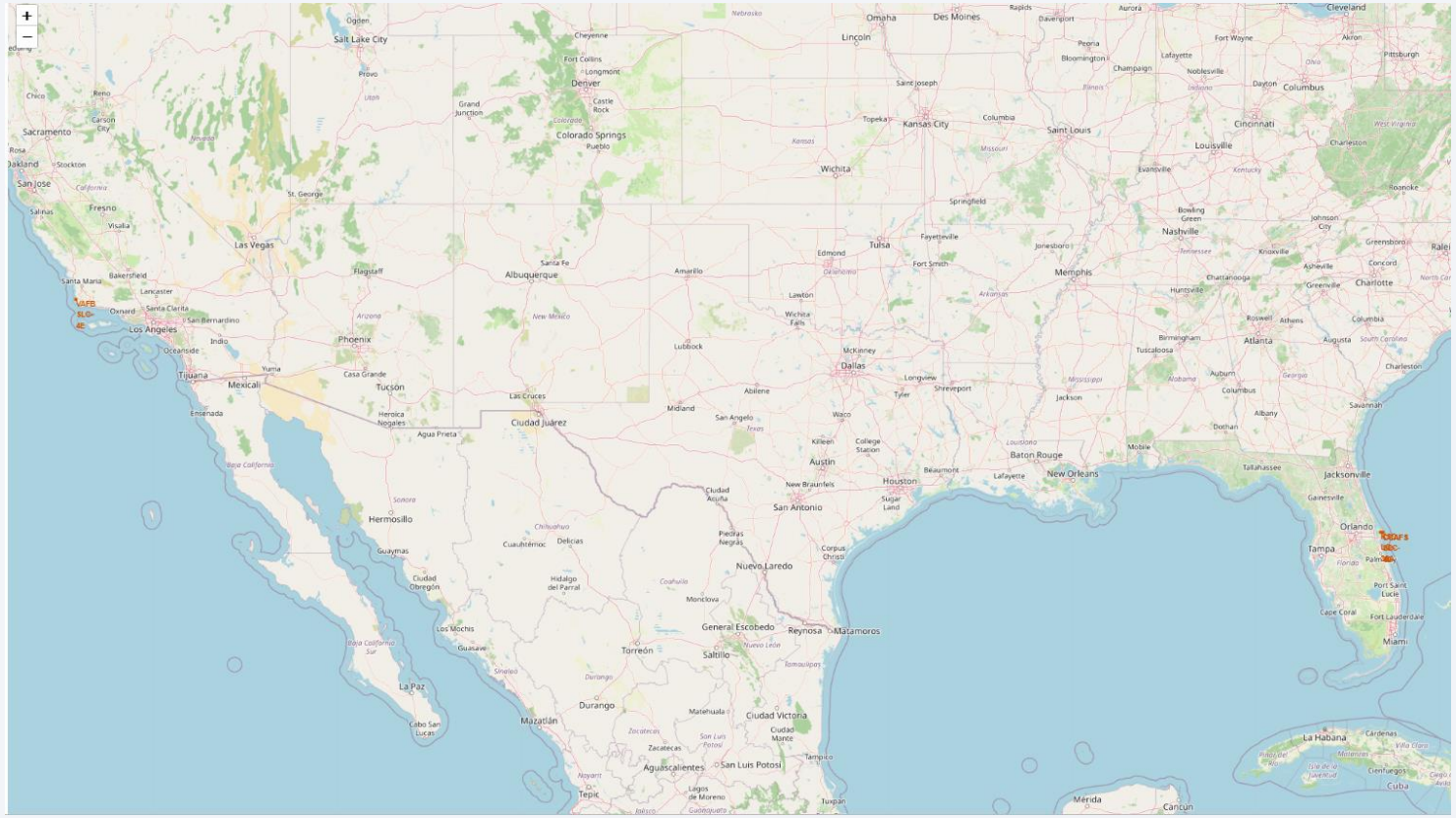
No attempt: 10, Success drone ship: 5, Failure drone ship: 5, Success ground pad: 3, Controlled ocean: 3, Uncontrolled ocean: 2, Failure parachute: 2, Precluded drone ship: 1

- Early launches show more uncontrolled outcomes — SpaceX was still developing landing technology during this period.

Section 3

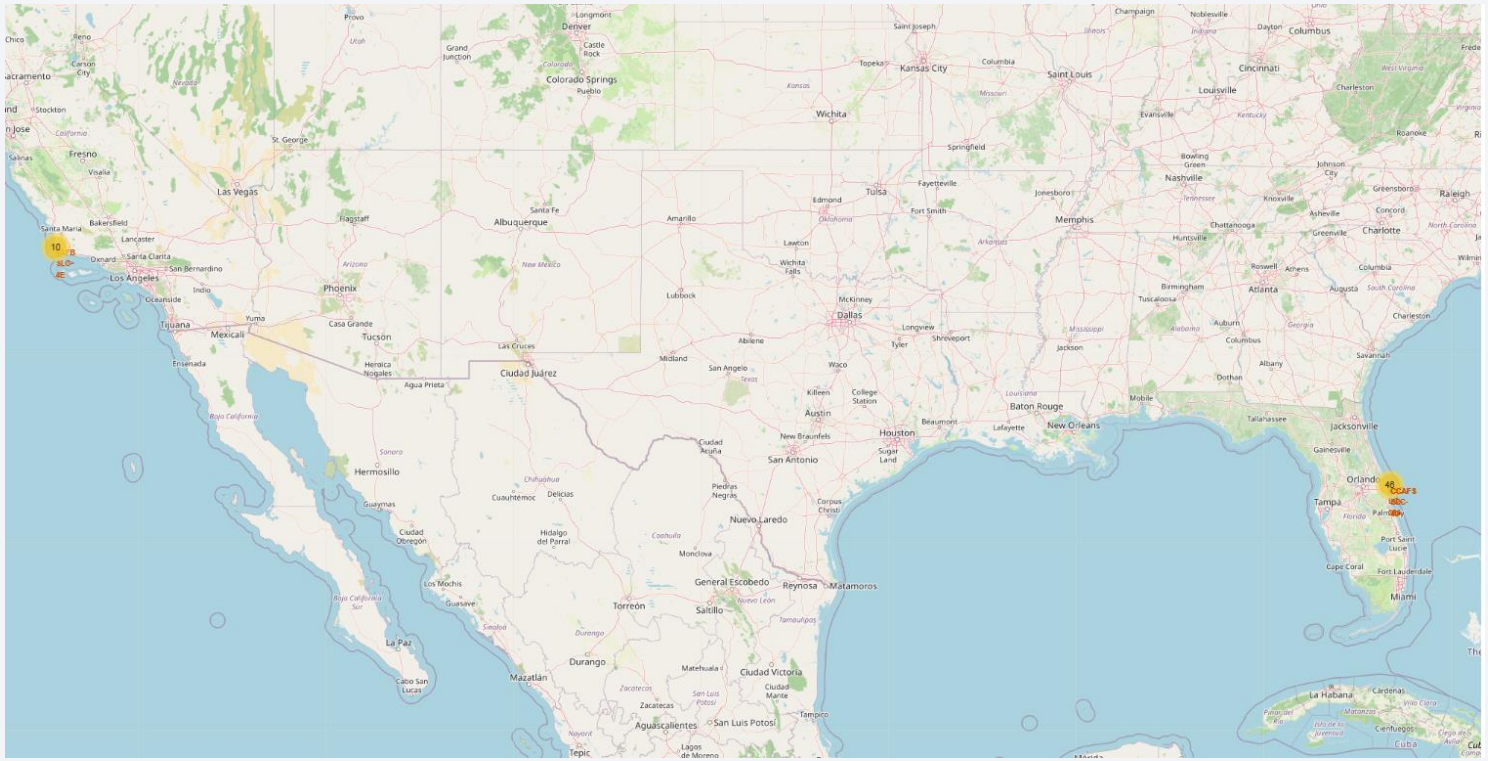
Launch Sites Proximities Analysis

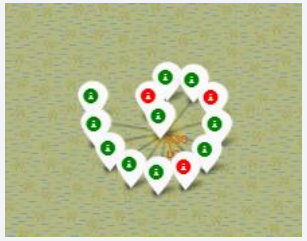
Falcon 9 Launch Sites — Global Location Map



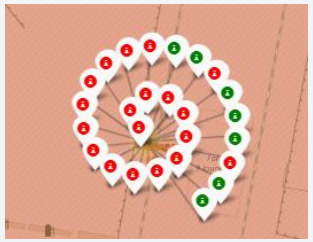
- Explanation: Four launch sites marked with circles and labels: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A (Florida), and VAFB SLC-4E (California). All sites are coastal, confirming geographic pattern of proximity to open water.

Launch Outcomes by Site — Success and Failure Distribution





(example: KSC LC-39A)

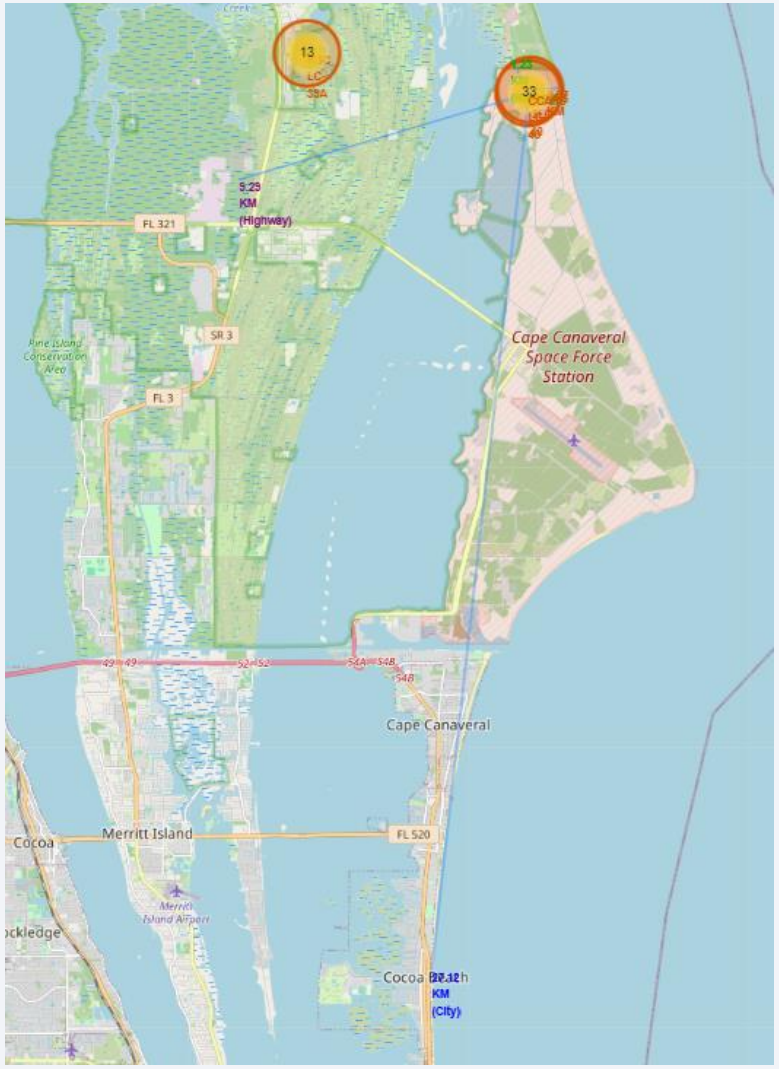


(example: CCAFS SLC-40)

- Explanation: Color-coded markers show individual launch outcomes: green = successful landing, red = failed. MarkerCluster groups nearby launches. KSC LC-39A shows highest concentration of green markers, confirming highest success rate.

CCAFS Launch Site — Proximity Analysis

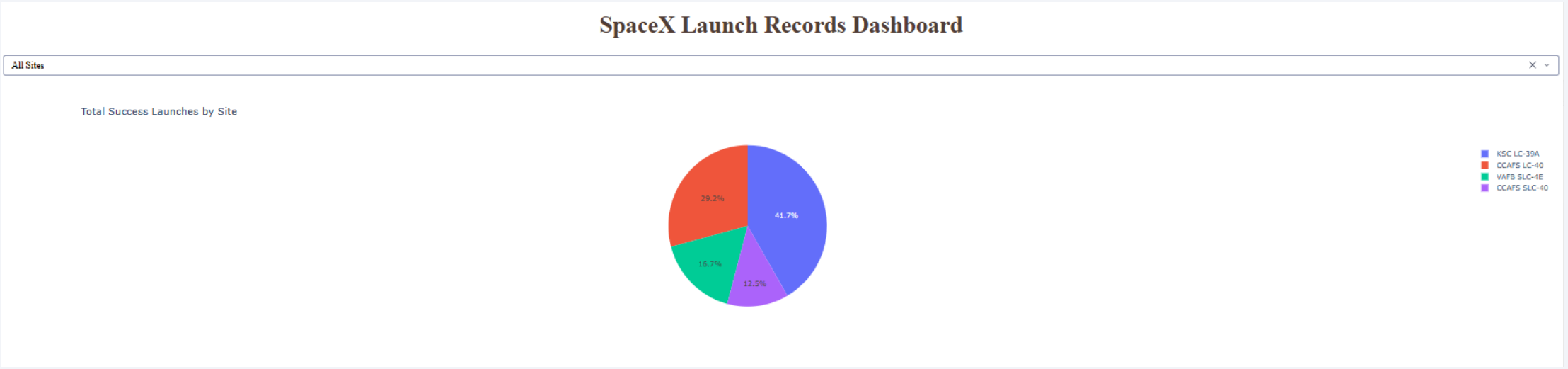
- Explanation: Distance lines drawn from CCAFS to nearest coastline (under 1 km), railway (1.23 km), highway (9.29 km), and city — Cocoa Beach (24+ km). Pattern confirms launch sites are deliberately positioned: coastal for safety, rail-accessible for logistics, buffered from population.



Section 4

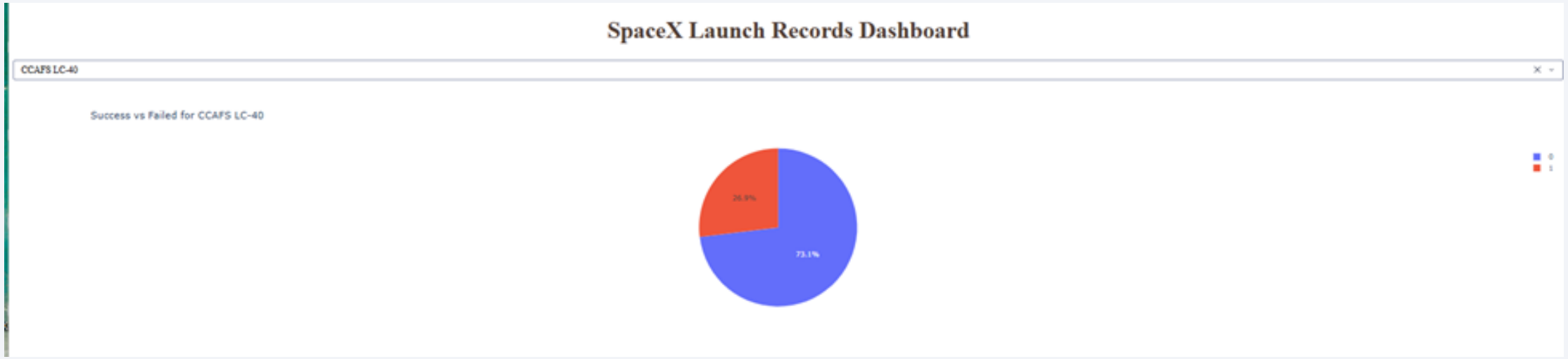
Build a Dashboard with Plotly Dash

Launch Success Count — All Sites



- Pie chart shows KSC LC-39A has the largest share of successful launches. CCAFS SLC-40 shows the most total launches but lower success ratio.

KSC LC-39A — Highest Success Rate Site



- KSC LC-39A pie chart shows 76.9% success rate — highest among all sites. Later flight numbers correlate with improved outcomes.

Payload vs Launch Outcome — All Sites

- Scatter plot reveals FT booster version has highest success concentration. Lower payload range (2000-4000 kg) shows highest success rate. Heavier payloads above 6000 kg show more mixed outcomes.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

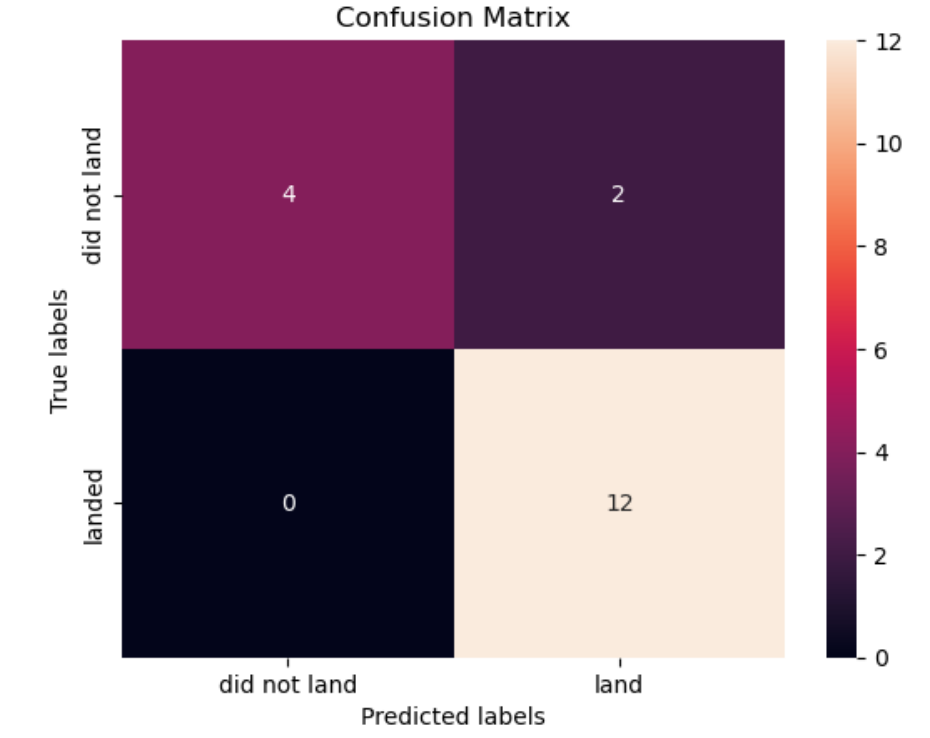
- The built model accuracy for all built classification models:

Model	*Test Accuracy*
Logistic Regression	83.3%
SVM	83.3%
KNN	83.3%
Decision Tree	88.9%

- The highest classification accuracy: Decision Tree

Confusion Matrix

- The confusion matrix of the best performing model: Decision Tree



The confusion matrix for the Decision Tree model shows: 12 true positives (correctly predicted landing), 4 true negatives (correctly predicted failure), 2 false positives (predicted landing but failed), and 0 false negatives (never missed an actual landing). Test accuracy: 88.9%. The model is particularly strong at identifying successful landings with no missed cases.

Conclusions

- KSC LC-39A has the highest launch success rate (76.9%) and the most successful launches overall
- Launch success improved significantly over time — from low rates in 2013 to near 100% by 2020
- Lower payload ranges (2000–4000 kg) correlate with higher landing success rates
- All launch sites share the same geographic pattern: coastal location, rail access, distance from cities — deliberate design for safety and logistics
- Decision Tree classifier achieved the highest test accuracy (88.9%) among four models tested, making it the best predictor of Falcon 9 first stage landing success

Appendix

- *All Python code, SQL queries, notebook outputs, and datasets are available in the GitHub repository: https://github.com/KatyaMak/IBM_Capstone*

Thank you!